

Security Operations Center (SOC) Incident Report

Brute Force Attack Detection – RDP (Hydra Simulation)

1. Executive Summary

This report documents a simulated brute force attack conducted against a Windows host using Hydra from a Kali Linux machine. The objective was to validate detection capabilities within a SIEM environment using Splunk. The attack targeted RDP and generated multiple failed authentication attempts. Logging and alerting mechanisms successfully captured the activity.

2. Environment Overview

Attacker Machine: Kali Linux VM
Target Machine: Windows VM (RDP enabled)
Network: Same virtual network
SIEM Tool: Splunk
Attack Vector: RDP brute force

3. Attack Setup & Execution

Network connectivity was verified via ping. Advanced audit policies were enabled to log authentication events. RDP was enabled to expose the attack surface.

```
hydra -l administrator -P /usr/share/wordlists/rockyou.txt rdp://<target-ip>
```

The attack generated multiple failed authentication attempts.

4. Detection & Analysis (Splunk)

Relevant Event IDs:
4625 – Failed logon
4624 – Successful logon (not observed)
4740 – Account lockout (not observed)

```
index=windows_logs EventCode=4625  
| stats count by src_ip, user  
| sort -count
```

Indicators: High volume failed logins, repeated attempts from single IP, targeting Administrator account.

5. Findings

No Event ID 4624 was observed, indicating no successful compromise. No Event ID 4740 suggests lockout policy was not triggered or not configured.

6. Recommendations

Enforce strong passwords, enable account lockout policies, restrict RDP access, implement MFA, and monitor failed login patterns.

7. Conclusion

The simulation demonstrated effective detection of brute force activity using Splunk. Proper logging enabled visibility into attack patterns.